



Sultan Qaboos University
College of Engineering
Office of Assistant Dean for
Industrial Training and Community Services

Weekly Report

Week #: 2

Date: 10/07/2026

Student	Full Name: Malek Mohammed Khamis Al Shukily	ID: 141776
	Department: ☐ CAE ☐ ECE ☐ MIE ☐ PCE ☒ MCE	Mobile: 72471416
Supervisor	Full Name: SATU Varuhoic	Email:
	Training Company & Department: MODES	Tel./Mobile: 335314

Tasks Completed (Briefly Describe any tasks that you completed during this week):

- Got hands-on with the workshop machines and carbon steel tool heads.
- Trained on Precision measuring tools: Vernier caliper and Micrometer.

Tasks in Progress (if any):

- Continuing to practice safe and accurate use of the Vernier caliper and Micrometer on different workpieces to build measurement confidence and speed.

Plan for next week (if different from this week):

- We will move to the large fuel facilities section, to be introduced to the equipment used there and the relevant safety procedures.

Problems/Challenges/Recommendations (if any):

No major problem phase us this week.

Supervisor Comments (your feedback is very important for our continuing improvement):

Satisfactory (Good Progress). Has a solid understanding of basic electrical and mechanical; continues to improve on hands. Work safely and maintains a clean workspace which reflects positively in the shop. Grasps the function of major components with more practice: specially electrical and mechanical.

Student:

I have read and understood the whole content of the Students' Training Manual.

Signature:

Supervisor:

I will do my best to guide the student towards achieving all required deliverables.

Signature:

Note: A copy of this report should be faxed to 24413416 or scanned and emailed to your SQU supervisor on a weekly basis. Originals should be placed in the Appendix of the **Final Report**.

WEEKLY WORKSHOP TRAINING REPORT

Department of Manufacturing Engineering

Student Name: Malek Al Shukily

Academic Week: Week 2 (Introduction & Workshop Familiarization)

Date of Report: July 10, 2026

1. Executive Summary & Objective

This weekly report details the educational and hands-on activities undertaken as a student inside the Manufacturing Engineering Department workshop. The primary focus of this week was to gain close-up familiarization with heavy industrial manufacturing machinery, observe their practical operations, understand the integration of specialized cutting tool materials like carbon steel inserts, and practice precision metrology. Furthermore, comprehensive safety regulations and fundamental mechanical cutting mechanisms were introduced.

2. Industrial Machinery Familiarization & Close-up Observation

During this training period, direct access to premium industrial-grade manufacturing equipment allowed for an in-depth understanding of workshop mechanics. The key machines inspected and studied included:

- **Colchester Student 1800 Lathe Machine:** A highly precise and robust conventional lathe used for various turning, facing, threading, and knurling operations. The alignment of the headstock, carriage, tool post, and tailstock was analyzed in detail.
- **Richmond Herald & Richmond SR2 Radial Drilling Machines:** Heavy-duty radial arm drills designed for cutting holes in large and dense workpieces. The mechanisms allowing the spindle head to move along the arm and the arm to rotate/elevate on the main column were demonstrated.
- **Behringer H.2 & Industrial Power Hacksaws:** Reciprocating power sawing systems utilized for efficiently slicing heavy steel bars and raw stock material to specified preliminary lengths before machining operations.
- **Industrial Horizontal Bandsaws:** Continuous band cutting machinery delivering smooth, high-precision, and continuous straight cuts on dense metallic profiles with integrated coolant fluid systems.

3. Tooling and Tooling Materials (Carbon Steel High-Performance Tips)

A vital component of the weekly instruction centered around tool geometry and material science. We explored the implementation of Carbon Steel (High-Carbon Steel) cutting tips and cutting edges. While high-carbon steel contains between 0.6% and 1.5% carbon, which grants it remarkable hardness and wear resistance at lower cutting temperatures, it requires precise operational speeds. We observed how carbon steel tips are effectively employed in low-speed cutting operations, specific wood/metal saws, and customized hand tools because they can maintain an exceptionally sharp cutting edge compared to other alloys.

4. Implementation of Precision Metrology Instruments

Quality control and dimension validation are essential in manufacturing. We received intensive technical training on how to handle, calibrate, and correctly read engineering measurement tools to determine external/internal diameters, depth, and linear lengths:

- **Vernier Caliper:** Utilized for verifying outer diameters (OD), inner diameters (ID) of bored cylinders, and overall component lengths. We learned to combine the main scale readings with the fractional vernier scale to achieve accuracy down to 0.02 mm.
- **Outside Micrometer:** Engineered for ultra-precise measurement of stock diameters and thicknesses. Using the ratchet stop mechanism to ensure uniform measuring pressure, we achieved highly consistent metric readings with a standard resolution of 0.01 mm.

5. Key Takeaways and Future Action Plan

The initial week in the manufacturing workshop successfully bridged theoretical engineering concepts with real-world shop-floor execution. Key learning outcomes include mastery over precision tool readings, recognition of optimal machine tools for distinct stock preparation, and understanding tool wear properties of high-carbon steel tips. In the upcoming week, training will transition from observation to practical machine setup, material loading, and initial turning operations on the lathe.